$$P_{10} = \sqrt{\frac{\sum p_0 q_1}{\sum p_1 q_1} \times \frac{\sum p_0 q_0}{\sum p_1 q_0}}$$

$$P_{01} \times P_{10} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1} \times \frac{\sum p_0 q_1}{\sum p_1 q_1} \times \frac{\sum p_0 q_0}{\sum p_1 q_0}}$$

Hence, $P_{01} \times P_{10} = \sqrt{1} = 1$

Factor reversal test

This test is also suggested by Fisher According to the factor reversal test, the product of price index and quantity index should be equal to the corresponding value index.

In Fisher's words "Just as each formula should permit the interchange of two times without giving inconsistent results so it ought to permit interchanging the prices and quantities without giving inconsistent results. i.e, the two results multiplied together should give the true ratio".

This statement is expressed as follows:

$$P_{01} \times Q_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$$

$$\text{Now, } P_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}}$$

$$Q_{01} = \sqrt{\frac{\sum p_0 q_1}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_1 q_0}}$$

$$\text{Hence, } P_{01} \times Q_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1} \times \frac{\sum p_0 q_1}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_1 q_0}}$$

$$= \sqrt{\frac{\sum p_1 q_1}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_0}}$$

$$= \frac{\sum p_1 q_1}{\sum p_0 q_0}$$

Circular Test

It is an extension of time reversal test. The time reversal test takes into account only two years. The current and base years. The circular test would require this property to holdgood for any two years. An index number is said to satisfy the circular test when there are three indices, P_{01}, P_{12} and P_{20}, such that $P_{01} \times P_{12} \times P_{20} = 1$.

Laspeyres, Paasche's and Fisher's ideal index numbers do not satisfy this test.

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